=== Hercules (NOAA/MSU) ===

Run JEDI (FV3-JEDI)

Note) If you have already set up the repository and input dataset on Day 1, please skip the first three steps (1 - 3).

1. Clone the following repository:

```
git clone https://github.com/NOAA-EPIC/CADRE-DA-training
```

2. Build the GDAS App (optional):

```
cd CADRE-DA-training/year2_cases

./build_gdas.sh
```

Note) The building process takes about 20 minutes on Hercules.

3. Check if the JEDI executables ('fv3jedi_var.x' and 'gdas_fv3jedi_fv3inc.x') exist in the following path (optional):

```
cd ../GDASApp

# to check the build status:

tail -n 10 build.log

# check executables in bin dir:

cd build/bin    (This is "JEDI_BIN_PATH" in the job submission script.)

ls

cd ../../../CADRE-DA-training/year2_cases
```

4. Copy the JEDI input yaml files to the specific location:

```
cd input_yaml

cp Day1/jedi_3dvar* .

(or)

cp Day2/[exp_case]/jedi_3dvar* .

(or)

cp Day3/[exp_case]/jedi_3dvar* .
```

5. Open the job submission script and modify it as needed:

```
cd ..

vim run_3dvar_hercules.sh

(Check your account, JEDI_BIN_PATH, EXP_NAME_BASE)

:wq
```

6. Submit your job:

```
sbatch run_3dvar_hercules.sh
```

7. Check your job:

```
squeue -u (your username)
```

Note) Remember the job ID of your current job.

8. Move to the experimental case directory for your job:

```
cd exp_case/[EXP_NAME_BASE].[job_ID]
```

9. Check the log files:

```
vim OUTPUT.fv3jedi

Vim errfile_fv3jedi
```

10. Move to the output directory:

```
cd output
ls
```

```
(plot_pyenv) ubuntu@ip-10-29-93-229:~/cadre26_noaa_tutorial/exp_case/cadre26.132/output$ ls
cubed_sphere_grid_atminc.jedi.nc          obs.20240224.t00z.atms_n20.satbias_cov.nc
diag.atms_n20_2024022400.nc               ufsda.t00z.atminc.cubed_sphere_grid.tile1.nc
diag.conventional_ps_2024022400.nc        ufsda.t00z.atminc.cubed_sphere_grid.tile2.nc
diag.gnssro_cosmic2_2024022400.nc         ufsda.t00z.atminc.cubed_sphere_grid.tile3.nc
diag.satwnd.abi_goes-16_2024022400.nc     ufsda.t00z.atminc.cubed_sphere_grid.tile4.nc
diag.scatwnd.ascat_metop-b_2024022400.nc  ufsda.t00z.atminc.cubed_sphere_grid.tile5.nc
obs.20240224.t00z.atms_n20.satbias.nc     ufsda.t00z.atminc.cubed_sphere_grid.tile6.nc
```

Run Diagnostics

1. Set up the diagnostic toolkit:

Option 1) Use the pre-installed diagnostic toolkit on Hercules:

```
source /work2/noaa/epic-explorer/cadre2026/hercules.anaconda
```

Option 2) Install the diagnostic toolkit directly:

```
cd CADRE-DA-training

git clone https://github.com/jkbk2004/ufs_da_diagnostics

# The following pip command only needs to run once

cd ufs_da_diagnostics

module load miniconda3

pip install --upgrade pip

pip install -e .

pip install seaborn scipy
```

Note) If you have set up the toolkit on Day 1, please skip the above steps.

```
export PATH=$HOME/.local/bin:$PATH
```

2. Update the input YAML files for the diagnostic tools:

```
cd ../diagnostic/yamls/[exp_case]

vim increment_maps.yaml

# change “prefix”s under “experiments”

:wq

vim obs_diag.yaml

# change “diag”s to the paths of your output

:wq

vim spectra_bkg_inc.yaml

# change “atm_file” under “background” and “prefix” under “increments”

:wq
```

```
vim spectra_ana_inc.yaml

# change “prefix”s under “experiments”

:wq
```

3. Create the “test” directory:

```
cd ..
mkdir -p test
cd test
```

4. Run the diagnostic tools one by one:

Day1:

```
ufsda-inc-maps  --yaml ../yamls/day1/increment_maps.yaml

ufsda-obs-diag  --yaml ../yamls/day1/obs_diag.yaml

ufsda-spectra-bkg-inc  --yaml ../yamls/day1/spectra_bkg_inc.yaml

ufsda-jedi-log ../../year2_cases/exp_case/[EXP_NAME_BASE].[job_ID]/OUTPUT.fv3jedi --output

day1_log_report.txt
```

Day2 (nicas):

```
ufsda-inc-maps  --yaml ../yamls/day2_nicas_length_scale/increment_maps.yaml

ufsda-obs-diag  --yaml ../yamls/day2_nicas_length_scale/obs_diag.yaml

ufsda-spectra-ana-inc  --yaml ../yamls/day2_nicas_length_scale/spectra_ana_inc.yaml

ufsda-jedi-log ../../year2_cases/exp_case/[EXP_NAME_BASE].[job_ID]/OUTPUT.fv3jedi --output

day2_log_report.txt
```

Day3 (error):

```
ufsda-inc-maps  --yaml ../yamls/day3_atms_err/increment_maps.yaml

ufsda-obs-diag  --yaml ../yamls/day3_atms_err/obs_diag.yaml

ufsda-spectra-ana-inc  --yaml ../yamls/day3_atms_err/spectra_ana_inc.yaml

ufsda-jedi-log ../../year2_cases/exp_case/[EXP_NAME_BASE].[job_ID]/OUTPUT.fv3jedi --output
day3_log_report.txt
```

Download File to Your Laptop

Open a new terminal window:

File:

```
scp [user_name]@hercules-dtn.hpc.msstate.edu:[path to the file] .
```

Directory:

```
scp -r [user_name]@hercules-dtn.hpc.msstate.edu:[path to the file] .
```